

Question 11 – Energy Levels and Emission Spectra

(a)(i) Determine the number of possible emission lines

✓ 1 mark

To calculate the number of different photon emissions (transitions) between 5 energy levels:

$$\text{Number of lines} = \frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$$

✓ Each transition emits a photon → each pair of levels produces one line.

(a)(ii) Identify the transition that gives the shortest wavelength

✓ 1 mark

- The **shortest wavelength** corresponds to the **greatest energy difference**
- From the diagram:

$$E_0 = -21.8 \times 10^{-19} \text{ J}, \quad E_3 = -1.36 \times 10^{-19} \text{ J}$$

$$\Delta E = E_3 - E_0 = (-1.36 - (-21.8)) = 20.44 \times 10^{-19} \text{ J}$$

✓ **Answer:** The **E3 → E0** transition produces the **shortest wavelength** photon.

(b)(i) Calculate the wavelength of a photon emitted in the E3 → E1 transition

✓ 3 marks

Given:

$$\bullet \quad E_3 = -1.36 \times 10^{-19} \text{ J}$$

$$\bullet \quad E_1 = -5.45 \times 10^{-19} \text{ J}$$

$$\Delta E = E_3 - E_1 = (-1.36 - (-5.45)) = 4.09 \times 10^{-19} \text{ J}$$

Use:

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$$

$$\lambda = \frac{6.63 \times 10^{-34} \cdot 3.00 \times 10^8}{4.09 \times 10^{-19}} = 4.86 \times 10^{-7} \text{ m} = 486 \text{ nm}$$

✓ This matches the **blue-green** hydrogen line in the data sheet.

 Revision Tips

- More energy lost = shorter wavelength = higher frequency photon
- Use:

$$E = hf = \frac{hc}{\lambda}$$

- To find total emission lines from levels:

$$\frac{n(n-1)}{2}$$

- Energy level diagrams can model real atomic spectra — the values link  exactly to observed light lines